P1-14 — Mathematical Advancement Without L1 (Independent Audit, New Corollaries, Moment Re-derivation)

Command: P1-14 — Mathematical Advancement Without L1 **Date:** 21 October 2026

Repository state (verified before any change): branch arena/ab8e330d-galios-hull, local HEAD = remote = 20423cfe22f279728cc380b925da47801b1384fa ("P1-12: L1 follow-up"), working tree clean. Frozen blueprint 02_THEOREM_BLUEPRINT.md SHA-256 ec2fd8d58f8d2a986314f83997d750d689f7c88e98ec8f87753ab4c9cf438cff — **not edited**, and no proof file or validation script was edited. This file, code/validate_p1_14.py, code/validation_p1_14_output.txt, and P1_14_PROPOSED_MANUSCRIPT_STRUCTURE.md are new artifacts.

What "independent" means here. code/validate_p1_14.py imports **none** of the earlier validation scripts: it re-implements finite fields GF(p^d), polynomial factorization, τ-orbits, the transfer trace, closed-walk counts, and Gram-rank hull dimensions from scratch, and it checks the formulas by three disjoint computational paths (direct exponent-tuple enumeration; fresh polynomial matrix powers; integer-evaluation matrix traces), plus a symbolic (sympy) re-derivation layer. Earlier validators were also re-run unmodified as a baseline (all pass; validate_p1_03.py output byte-identical to its saved output). **Computational agreement is evidence, never proof**; every result below is labelled accordingly.

Headline outcome. No counterexample to any frozen statement (T1–T6, corrected C1–C6) was found — in the proofs (re-read line by line), in a fresh symbolic re-derivation of every moment closed form, or in exact enumeration over 52 orbit configurations and 2,769 field-level assignments. Three **new corollaries (N1–N3)** are proved completely in §4; they are **proposals** (the frozen blueprint is untouched; adoption needs a Commander decision, exactly like the earlier PC items). L1 and Gate G1 are untouched: **no novelty or priority claim is made anywhere in this file.**

---

1. Theorem-by-theorem audit (T1–T6, C1–C7, and the blueprint §2 identity)

Each statement was re-read in 02_THEOREM_BLUEPRINT.md and checked against its proof in P1_04_PROOFS_T1_T4.md, P1_05_PROOFS_T5_T6_COROLLARIES.md, P1_06_MOMENT_PROOFS_C5_C6.md, then re-tested by the fresh code. Verdicts:

Statement | Audit method | Verdict

T1 (assignments ⟷ codes; `(P+1)^{ | F_s | }`) | Proof re-read (RR-1 via PID/UFD Facts 0.1–0.2). Fresh field cases reproduce totals `(P+1)^{ | F | }` exactly (27, 27, 256, 256, 16, 2187). | **VERIFIED**

T2 (# multiplicative; root transport; compatibility two forms) | Proof re-read (RR-2 (i)–(vi)). Domain discipline of PC-3 respected (f(0) ≠ 0, automatic since S(0) = −μ_s ≠ 0). Fresh code asserts τ permutes the factor set in every field case. | **VERIFIED** (within its stated domain)

T3 (dual exponents β_i = P − α_{i−1}) | Proof re-read (RR-3 → Prop. 4.2 → T3). The code-first slot order (S7) is exactly what produces ρ = σ⁻¹ and α_{i−1}; the opposite reading would give the σ-version and α_{i+1}. Fresh Gram-rank brute force pins the orientation **per assignment** (see §2.3). | **VERIFIED** under the frozen code-first convention

T4 (hull generator lcm; dimension sums) | Proof re-read (RR-5 + T3). Fresh check: for **every** assignment in six field cases (2,769 total), (dim C, dim Hull) from the Gram matrix equals the T4 formula. | **VERIFIED** (compatible case)

T5 (transfer trace identity) | Proof re-read (closed-walk expansion). Fresh: polynomial trace == direct enumeration == integer-evaluation trace on the grid of §2.1. | **VERIFIED**

T6 (global joint enumerator) | Proof re-read (Prop. 3.1 assembly + distributive law). Layer D rebuilds a two-component global distribution by exact convolution of fresh orbit distributions and reproduces P1-05's published counts (729 / 16 / 144 / 144) with none of its code. | **VERIFIED**

C1 (total `(P+1)^{Σ | F_s | }`) | J-power argument re-read; totals match everywhere. | **VERIFIED**

C2 (code-dimension marginal) | Rank-one trace argument re-read; per-orbit z=1 marginal checked on the whole Layer-B grid. | **VERIFIED**

C3 (LCD count 2^{#orbits}) | Proof re-read (repaired boundary argument in P1-05 §5). [z^0] = 2 per orbit on the whole grid; field LCD counts 2^{#orbits} in all six cases. Strengthened as **N3** (§4). | **VERIFIED**

C4 (self-orthogonal / dual-containing, incl. PC-4 count) | Nonnegative-edge-difference argument re-read. Per-orbit and

global closed-walk counts tr(B_P^L), tr((B′_P)^L) match enumeration everywhere tested; the two counts are equal (involution α ↦ P−α), e.g. GF(9), λ=−1: 100 = 100. | **VERIFIED**

C5 (means, corrected split) | §5: symbolic re-derivation + enumeration + field cases with pure/mixed orbit lengths. The pre-PC-1 display is refuted concretely (§5.3). | **VERIFIED** (in its frozen, corrected form)

C6 (variance, three regimes) | §5: every closed form re-derived symbolically from tail sums (not from P1-06's hand algebra); enumeration grid + field variances match (e.g. GF(8): brute 10/9 = V_1(2) + V_6(2) = 2/9 + 8/9). | **VERIFIED**

C7 (EAQECC) | Conditional on an externally known distance; the enumerator alone gives no distance. Not a proved enumerator consequence, by design. | **OUT OF SCOPE — conditional only; must never be stated as following from the enumerator**

Blueprint §2 layer identity min(P−α_i, α_{i−1}) = Σ_t ε_{i−1}^{(t)}(1−ε_i^{(P−t+1)}) | Proved here (§2.4) and checked for P = 1..9, all exponent pairs. | **VERIFIED**

No frozen statement requires correction. The genuine defects found are documentation/reproducibility issues; they are listed honestly in §7.

---

2. Independent audit of the transfer-matrix enumerator (T5/T6)

The audited formula (frozen T5/T6): E(u,z) = ∏_s ∏_O tr((T^{(P)}_{w_O}(u,z))^{L_O}), with T_{a,b} = u^{w(P−b)} z^{w·min(P−b, a)}, row a = previous exponent, column b = current exponent.

2.1 Three disjoint computational paths, one grid

For P ∈ {1,2,3,4,5,7,8,9}, all L ≥ 1 with (P+1)^L ≤ 100000 (52 (P,L) pairs; w = 1, plus w ∈ {2,3} spot configurations — 58 orbit configurations with moments checked), the fresh script compared:

1. **Ground truth:** direct enumeration of all α ∈ {0,...,P}^L, accumulating (wΣ(P−α_i), wΣ min(P−α_i, α_{i−1}));
2. **Polynomial trace:** tr(T^L) with bivariate-polynomial matrix entries (fresh implementation);
3. **Integer-evaluation trace:** entries replaced by integers at $(u_0,z_0) \in \{(2,3),(3,5)\}$, ordinary integer matrix power, trace compared against the enumeration polynomial evaluated at the same points.

All three agree on the full joint distribution in every configuration. This is strong evidence; the proof itself is the closed-walk expansion (P1-05 §2), re-read and found complete: the bijection between closed walks a_0 → ... → a_{L−1} → a_0 and assignments is exact **because orbit positions are labeled factors** — there is no division by L (a walk and its rotation are different assignments), and the totals confirm it: the trace sums to exactly (P+1)^L, never to (P+1)^L / L.

2.2 Indexing, edge cases, coefficient interpretation

- **Indexing.** Row = predecessor exponent a_{j−1}, column = current a_j; the edge weight min(P−a_j, a_{j−1}) is exactly T4's hull term at position j. The closing edge a_{L−1} → a_0 supplies the term at position 0 with predecessor α_{L−1} (indices mod L). For L = 1 this degenerates correctly to the self-loop tr(T) = Σ_a u^{w(P−a)} z^{w·min(P−a,a)}, matching α_{−1} = α_0.
- **P = 1 edge case.** T^{(1)}_w(u,z) = [[u^w, 1],[u^w z^w, 1]] (rows/cols 0,1), the predecessor binary matrix of P1-05 §6; at P=1 the hull edge term is the 1→0 transition indicator. The grid includes P=1, L=1..8.
- **Coefficient interpretation.** The coefficient of u^K z^H in tr(T^L) is the number of exponent tuples on that orbit with code-dimension contribution K and hull contribution H (both in F_q-units after the weight w). Verified coefficient-wise (full distribution equality, not just moments). Coefficients are nonnegative integers; E(u,z) ∈ Z[u,z] as claimed.
- **Weight scaling.** T^{(P)}_w(u,z) = T^{(P)}_1(u^w, z^w) entrywise, so tr((T_w)^L)(u,z) = tr((T_1)^L)(u^w, z^w); the w ∈ {2,3} checks confirm the scaling behaves as the formula predicts.
- **Specializations.** u=z=1: (P+1)^L per orbit (C1 mechanism). z=1: per-orbit code marginal (Σ_{j=0}^P u^{wj})^L (C2 mechanism) — checked on the grid. u=1: hull marginal (no product

closed form exists in general — this is precisely why the transfer matrix is needed). [z^0] = 2 per orbit (C3 mechanism).

- **Hypothesis separation (a clarification the audit makes precise).** As an identity in Z[u,z], T5 holds for **every integer P ≥ 1** — the proof is pure matrix multiplication; the prime-power condition P = p^ϱ is needed only upstream, where T1/T4 realize orbits and exponents from an actual field factorization. The audit's P ∈ {5,7} grid points exercise the combinatorial identity outside the repeated-root field construction; no theorem statement is thereby extended.

2.3 The one thing computation alone cannot see: orientation

Replacing T by its transpose produces the distribution of Σ_i min(P−α_i, α_{i+1}), which equals the audited distribution under the reversal bijection α_i ↦ α_{−i}. **Hence no trace/distribution computation can distinguish predecessor from successor.** The frozen α_{i−1} (predecessor) is fixed by the proof chain T3 (β_i = P − α_{i−1}, from the code-first pairing) → T4, and it is pinned empirically where it matters: the fresh GF(8) case (k=1, λ=1, P=2, n_0=7, orbit lengths [1,6]) compares the T4 formula against Gram-rank hull dimensions **assignment by assignment** over all 2,187 assignments — on an L=6 orbit, predecessor vs successor give genuinely different hull dimensions for generic α, so this check (unlike any distribution check) certifies the orientation. (Orbits of length L ≤ 2 cannot distinguish the two readings; L=6 can and does.)

2.4 The blueprint §2 layer identity (used nowhere in the proofs, now verified)

For t ∈ {1,...,P}, ε^{(t)} = 1[α ≥ t]. With a = α_{i−1}, b = α_i: ε_{i−1}^{(t)} = 1 ⟺ t ≤ a and 1 − ε_i^{(P−t+1)} = 1 ⟺ t ≤ P−b. Hence Σ_{t=1}^P ε_{i−1}^{(t)}(1−ε_i^{(P−t+1)}) = #{t : t ≤ min(a, P−b)} = min(P−α_i, α_{i−1}). ∎ (Also checked for P=1..9, all pairs, in Layer B.) The optional Phase-4 layer-decomposition route therefore rests on a correct identity, should it ever be opened.

---

3. Corollaries re-derived from the verified theorem

C1–C4 were re-derived in full in P1-05 §5; this audit re-read those derivations and re-tested every conclusion computationally (§1 table). Two consequences are recorded here in the exact form the manuscript will need (frozen wording preserved; nothing new is claimed beyond what the cited proofs give):

- **Total-count identity (C1).** E(1,1) = (P+1)^{Σ_s |F_s|}, equivalently the labeled-assignment count; field totals 3^3, 3^3, 4^4, 4^4, 4^2, 3^7 reproduced exactly.
- **Marginals.** Code-dimension marginal is C2 (E(u,1) = ∏∏_f (1 + u^{w_f} + ⋯ + u^{Pw_f})). Hull-dimension marginal is E(1,z) = ∏_s∏_O tr((T^{(P)}_{w_O}(1,z))^{L_O}) with T(1,z)_{a,b} = z^{w·min(P−b,a)} — a trace product, **not** a factor product; its recorded special values are [z^0] = 2^{#orbits} (C3) and the first two moments (C5/C6).
- **LCD specialization (C3).** #{a : dim Hull_{A,k} = 0} = 2^{#orbits}, independent of P; per orbit exactly the two constant tuples α ≡ 0, α ≡ P. Reproduced in every layer (grid [z^0]=2; field counts 8,4,16,4,2,4 against orbit counts 3,2,4,2,1,2).
- **Simple-root recovery (P=1, where the hypotheses permit).** μ_s = λ_s, squarefree divisors, binary assignments; hull edge = 1→0 transition indicator; the T6 product is the predecessor binary enumerator (P1-05 §6, re-read). Moments at P=1: E[Hull] = (1/4)·Σ_{L_O≥2} w_O L_O (since ν_1 = 0), V_1(1) = 0, V_2(1) = 1/4, V_L(1) = L/16 — all inside the Layer-B grid.

---

4. New corollaries proved here (PROPOSED — not in the frozen blueprint)

Standing model (same as frozen): (G1)–(G6) of P1-05; uniform distribution on labeled assignments; per orbit O, A_i iid uniform on {0,...,P}, X_i = min(P−A_i, A_{i−1}) (indices mod L = L_O), orbit code contribution w_O Σ_i (P−A_i), hull contribution w_O Σ_i X_i. F(c) := E[min(c,U)] = c(2P−c+1)/(2(P+1)) (P1-06 Lemma 1.3) and μ_P = P(2P+1)/(6(P+1)) (P1-06 Lemma 1.1) are used.

N1. dim C and dim Hull are uncorrelated (new)

Statement. Cov(dim_{F_q} C(a), dim_{F_q} Hull_{A,k}(C(a))) = 0 under the uniform labeled model — for every component/orbit configuration, every P ≥ 1. (They are **not** independent; see the remark.)

Lemma N1.1 (one shared variable, "self" pairing). For independent B, C uniform on {0,...,P}, Cov(B, min(B,C)) = P(P+2)/24 =: γ_P. Proof. Condition on B = b: E[min(b,C)] = F(b), so E[B·min(B,C)] = (1/(P+1)) Σ_b b·F(b) = (1/(2(P+1)^2)) Σ_b b²(2P+1−b). With Σ_{b=0}^P b² = P(P+1)(2P+1)/6 and Σ b³ = (P(P+1)/2)², the sum equals (P/2)·(5P²+5P+2)/12, hence E[B·min(B,C)] = P(5P²+5P+2)/(24(P+1)). Also E[B]·E[min(B,C)] = (P/2)μ_P = P(4P²+2P)/(24(P+1)). Subtracting: γ_P = P(P²+3P+2)/(24(P+1)) = P(P+2)/24. ∎

Proof of N1. Orbit blocks involve disjoint exponent variables, hence are independent (product model, P1-05 Prop. 3.1); cross-orbit covariance terms therefore vanish, and with K_O = Σ_i (P−A_i), S_O = Σ_j X_j it suffices to show Cov(K_O, S_O) = 0 for one orbit, since Cov(dim C, dim Hull) = Σ_{s,O} w_O²·Cov(K_O, S_O). Expand: Cov(K_O, S_O) = Σ_{i,j} Cov(P−A_i, X_j), where X_j depends only on (A_j, A_{j−1}).

- If i ∉ {j, j−1} (possible only for L ≥ 3): the two sides are functions of disjoint subfamilies of independent variables ⇒ covariance 0.
- If i = j: Cov(P−A_j, min(P−A_j, A_{j−1})) = γ_P by Lemma N1.1 (with B = P−A_j, C = A_{j−1}).
- If i = j−1 (requires L ≥ 2): Cov(P−A_{j−1}, min(P−A_j, A_{j−1})). Put C = A_{j−1}, B = P−A_j (independent uniforms): this is Cov(P−C, min(B,C)) = −Cov(C, min(B,C)) = −γ_P, because (B,C) exchangeable gives Cov(C, min(B,C)) = Cov(B, min(B,C)).
- L = 1: the single term is Cov(P−A, min(P−A, A)). With B = P−A uniform, min(B, P−B) is invariant under B ↦ P−B, so E[B·min(B,P−B)] = E[(P−B)·min(B,P−B)]; averaging gives E[B·min] = (P/2)·E[min] = E[B]·E[min], i.e. covariance 0. (Both parity cases also confirmed symbolically, Layer A.)

For L ≥ 2 each j contributes exactly one γ_P (from i=j) and one −γ_P (from i=j−1) — for L=2 these two indices exhaust {0,1}, for L≥3 all other i contribute 0 — so Cov(K_O, S_O) = L·(γ_P − γ_P) = 0; for L=1 it is 0 directly. Summing with weights w_O² over all orbits and components gives 0. ∎

Remark (uncorrelated ≠ independent). At P=1, L=2, w=1: Pr(dim C = 2) = 1/4, Pr(dim Hull = 1) = 1/2, but Pr(dim C = 2, dim Hull = 1) = 0 ≠ 1/8 (fresh enumeration, Layer B). Any manuscript wording must say "uncorrelated", never "independent".

Status. Proved completely above (the only inputs are P1-06 Lemmas 1.1/1.3 and the product model). Verified exactly: Cov = 0 in all 58 Layer-B orbit configurations and in all six Layer-C field distributions and the Layer-D global distribution. **Adoption into the blueprint/manuscript result list requires a Commander decision; the frozen file is unchanged.**

N2. Variance of the code dimension (new)

Statement. Var(dim_{F_q} C(a)) = [P(P+2)/12] · Σ_s Σ_{f∈F_s} w_f², where w_f = m_s·deg_{K_s} f. Proof. By T4, dim_{F_q} C = Σ_{s,f} w_f (P − A_f) with the A_f independent uniform on {0,...,P} (product model). Var(P − A_f) = Var(U) for U uniform on {0,...,P}: E[U] = P/2, E[U²] = P(2P+1)/6, so Var(U) = P(2P+1)/6 − P²/4 = P(P+2)/12. Independence ⇒ variances add with coefficients w_f². ∎ Consistent with C5's mean E[dim C] = (n/2)·Σ_s m_s. Verified on the whole Layer-B grid (per orbit: w²L·P(P+2)/12), all field cases (e.g. GF(8): Var(C) = 14/3 = 7·(2·4/12)), and Layer D (190/3). **PROPOSED** (same adoption status as N1).

N3. LCD refinement: joint (LCD, code-dimension) count (new)

Statement. Σ_{a : Hull(a) = 0} u^{dim_{F_q} C(a)} = ∏_{s} ∏_{O∈O_s} (1 + u^{P·w_O·L_O}). At u=1 this is C3. Proof. By the C3 argument (P1-05 §5), Hull(a) = 0 iff on every orbit α ≡ 0 or α ≡ P. The orbit's code-dimension contribution (T4) is w_O Σ_i (P−α_i), equal to P·w_O·L_O for α ≡ 0 and to 0 for α ≡ P. Hence the per-orbit generating polynomial over LCD-admissible tuples is 1 + u^{P w_O L_O}, and independence over orbits/components (T1, Prop. 3.1) gives the product. ∎ Verified wherever C3

was verified ([z^0] counts plus the Layer-D convolution). **PROPOSED** (same adoption status).

---

5. Moment formulas C5/C6: symbolic re-derivation, boundary cases, covariance map

5.1 Independent symbolic re-derivation (Layer A)

Every closed form of P1-06 was re-derived from its **defining tail/finite sums** with sympy (summation over a symbolic P; the L=1 formulas via the parity substitutions P = 2m, 2m+1 as polynomial sums) — i.e. independently of P1-06's hand simplifications:

Quantity | Defining sum | Closed form (P1-06) | Symbolic match
μ_P | Σ_{t=1}^P ((P−t+1)/(P+1))² | P(2P+1)/(6(P+1)) | ✓
E[min(U,V)²] | Σ_t (2t−1)((P−t+1)/(P+1))² | P(P²+P+1)/(6(P+1)) | ✓
Var(edge) | difference | P(P+2)(2P²+4P+3)/(36(P+1)²) | ✓
F(c) | Σ_{t=1}^c (P−t+1)/(P+1) | c(2P−c+1)/(2(P+1)) | ✓
E[X_iX_{i+1}] | (1/(P+1))Σ_b F(P−b)F(b) | P(P−1)(11P²+15P+6)/(120(P+1)²) | ✓
κ_P | difference | −P(P+2)(7P²+14P+9)/(360(P+1)²) | ✓
E[X_0X_1] (L=2) | Σ_{s,t≥1, s+t≤P} ((P−s−t+1)/(P+1))² | P²(P−1)/(12(P+1)) | ✓
Cov_{L=2} | difference | −P²(P+2)²/(36(P+1)²) | ✓
ν_P, E[X²], V_1 (L=1, both parities) | Σ_a min(P−a,a)^{1,2} split at P=2m / 2m+1 | P1-06 Prop. 2.2 / 3.1 | ✓ ✓ ✓
V_2, V_{L≥3} assemblies | 2Var + 2Cov; Var + 2κ_P per position | P1-06 Prop. 3.6 / 3.4 | ✓ ✓
γ_P (new, for N1) | (1/(P+1))Σ_b bF(b) − (P/2)μ_P | P(P+2)/24 | ✓
Var(U{0..P}) (for N2) | Σ b²/(P+1) − (P/2)² | P(P+2)/12 | ✓

The same formulas were then compared against exact rational enumeration (fractions.Fraction, no floats) on the Layer-B grid and against the field distributions: **no discrepancy anywhere. No formula fails; no correction to the frozen C5/C6 is needed or proposed.**

5.2 Boundary cases (all inside the verified grid)

- **P = 1:** ν_1 = 0 (the self-loop min(1−A,A) ≡ 0), V_1(1) = 0; L≥2 edges are Bernoulli(1/4): Var = 3/16, and V_L(1) = L/16, V_2(1) = 1/4.
- **L = 2 is not the L≥3 formula:** at P=2, true V_2 = 44/81, whereas substituting L=2 into V_{L≥3} gives 8/27 = 24/81. The two edges share **both** variables; P1-06's separate treatment is necessary (recomputed independently here).
- **Parity at L=1:** the even/odd branches of V_1 are different formulas but asymptotically agree (~ P²/48); no discontinuity in P is introduced — e.g. V_1(2) = 2/9, V_1(3) = 1/4, V_1(4) = 14/25 (grid values all matched enumeration; the closed forms themselves are the symbolic identities of §5.1).
- **ν_P < μ_P:** self-loop mean (~ P/4) is strictly below the two-independent-uniforms mean (~ P/3); this gap is exactly the PC-1 correction.

5.3 The refuted pre-correction mean, quantified (why PC-1 mattered)

In the GF(4), n_0=3, P=2 field case all three orbits have L=1: true E[dim Hull] = 1 = 3·ν_2. The pre-PC-1 display μ_P·Σ_O w_O L_O would give 3·μ_2 = 5/3 ≠ 1. In GF(9), λ=1, P=3 (four L=1 orbits): true mean 2 = 4ν_3; pre-correction 4μ_3 = 7/2 ≠ 2. The frozen (corrected) C5 gives the true values in both cases — confirmed against Gram-rank brute force, not just the exponent model.

5.4 Covariance-dependence map (what depends on what)

- **Within an orbit, L ≥ 3:** adjacent edges are negatively correlated: κ_P < 0 for all P ≥ 1; correlation κ_P/Var(edge): P=1: −1/3, P=2: −13/38, P=3: −19/55. Non-adjacent edge pairs (L ≥ 4) are independent (disjoint variable pairs) ⇒ covariance 0; at L=3 every pair is adjacent.
- **L = 2:** the single pair shares both variables; correlation Cov_{L=2}/Var(edge): P=1: −1/3, P=2: −8/19, P=3: −5/11 — different from the adjacent-L≥3 values at the same P≥2, as it must be.
- **Across orbits/components:** independent by the product model ⇒ all cross-covariances 0 (this

is what makes C6 a sum with weights w_O^2).

- **Code vs hull dimension:** covariance exactly 0 (N1) despite dependence (witness in §4). Correlation coefficient is therefore 0 whenever both variances are positive — a clean, quotable fact, provided "uncorrelated" is not upgraded to "independent".

---

6. Worked examples (exact; reproduced by code/validate_p1_14.py)

E1 — one orbit, P=2, L=2, w=1 (all 9 tuples). From Layer B (tuple $\alpha=(\alpha_0,\alpha_1) \mapsto (K,H) = (\Sigma(2-\alpha_i),\ \Sigma \min(2-\alpha_i, \alpha_{i-1}))$): (0,0)↦(4,0), (0,1)↦(3,1), (0,2)↦(2,2), (1,0)↦(3,1), (1,1)↦(2,2), (1,2)↦(1,1), (2,0)↦(2,2), (2,1)↦(1,1), (2,2)↦(0,0). Hence $E[H] = 10/9 = 2\mu_2$, $Var(H) = 44/81 = V_2(2)$, $Cov(K,H) = 0$, $[z^0]=2$ — the frozen L=2 variance (not the L≥3 formula, cf. §5.2) and N1 visible by hand.

E2 — simple-root orbit, P=1, L=3, w=1. Joint distribution {(0,0):1, (1,1):3, (2,1):3, (3,0):1}: hull =1 exactly on the six non-constant binary necklaces' tuples with a single 1→0 transition class, $E[H] = 3/4 = 3\mu_1$, $Var(H) = 3/16 = V_3(1)$.

E3 — field case GF(4), k=1, λ=1, P=2, n_0=5, n=10 (fresh implementation; factor degrees [1,2,2], orbit lengths [1,2], 27 assignments). Joint (dim_K C, dim_K Hull) distribution: {(0,0):1, (1,1):1, (2,0):1, (2,2):2, (3,3):2, (4,2):2, (4,4):3, (5,5):3, (6,2):2, (6,4):3, (7,3):2, (8,0):1, (8,2):2, (9,1):1, (10,0):1}. Checks: brute = T4 formula = T6 transfer; $E[H] = 23/9$ (orbit split: $1\cdot\nu_2 + 2\cdot2\cdot\mu_2 = 1/3 + 20/9$), $Var(H) = 194/81$, $Var(C) = 6$ (N2), $Cov = 0$ (N1), $LCD = 4 = 2^2$, self-orthogonal = dual-containing = 12. The RR-3 reproduction (below) also ran on this case.

E4 — field case GF(8), k=1, λ=1, P=2, n_0=7, n=14 (orbit lengths [1,6], 2,187 assignments): brute = formula = transfer; $E[H] = 11/3$, $Var(H) = 10/9 = V_1(2) + V_6(2) = 2/9 + 8/9$, $Var(C) = 14/3$ (N2), $Cov = 0$ (N1), LCD = 4, self-orthogonal = dual-containing = 258. This is the case that pins the predecessor orientation (§2.3).

E5 — two-component assembly without P1-05 code (Layer D). Convolving the fresh per-orbit distributions for P=2, orbits (L,w) = (1,1), (2,3), (1,2), (2,6) (the A = GF(2)×GF(4) example): total 729, LCD 16 = 2^4, self-orthogonal 144, dual-containing 144 (all as published by P1-05, reproduced independently), plus $E[H] = 11$ (C5 split), $Var(H) = 230/9$ (C6), $Var(C) = 190/3$ (N2), $Cov = 0$ (N1).

RR-3 reproduction (closing a lost-evidence gap). P1-04 §9's direct RR-3 check ran a script kept only in /tmp. Layer C re-creates it freshly: for **every** assignment in the GF(4) n_0=3 and n_0=5 cases, the shifts of the ordinary reciprocal $h^\natural$ are verified Euclidean-orthogonal to every generator row of C and number deg g (the inclusion + dimension argument of RR-3). Passed on all 54 assignments. The evidence is now a repo artifact (code/validate_p1_14.py), not a /tmp file.

---

7. Unresolved proof gaps and audit flags (honest list)

No mathematical gap in T1–T6 / C1–C6 was found, and no counterexample exists in the tested range; no change to the frozen core is needed, and none is made. The following flags are documentation, scope, or literature items — recorded, not silently fixed:

1. **(Closed by this command)** RR-3's original direct computational check existed only as /tmp/check_rr3.py (P1-04 §9) and was not reproducible from the repo. Reproduced freshly in Layer C (E3 note above).
2. **Evidence layering.** validate_p1_05.py imports validate_p1_03.py, so the P1-05 multi-component evidence was never code-independent of P1-03; validate_p1_06.py tests the orbit model at formula level. validate_p1_14.py is import-free and re-derives the same conclusions (including P1-05's 729/16/144/144) — the dependence is now mitigated, but readers should know the earlier scripts share code.
3. **Frozen-file bookkeeping lags the proofs.** In 02_THEOREM_BLUEPRINT.md §5, the RR-1…RR-9 checkboxes and the "Prototype-2" box remain unchecked although RR-1…RR-8 are discharged in P1-04…P1-06 and the F_9 / two-component evidence exists (P1-03/P1-05). This is a

stale checklist inside a frozen file; correcting it would require a Commander-approved dated note (GAP-17 rule) and is **not** done here.

4. **Orientation is proof-borne, not computation-borne** (§2.3): any future exposition must keep the code-first convention (S7) visibly attached to T3/T4, because the enumerator's distribution alone cannot detect a predecessor/successor swap.

5. **Scope boundaries (not gaps):** labeled assignments only (not equivalence/isometry classes); compatible twists only (incompatible: T3 survives in two-modulus form, T4 onward is not claimed); square-free affine A; uniform labeled model for all moments; C7 conditional on external distance data.

6. **Literature (untouched by this command):** L1 full text still unacquired; L1-Q1–Q6 unresolved; **Gate G1 remains DEFERRED**. The mathematical strengthening in this file does not bear on novelty or priority.

---

8. What is proved vs what is computational evidence

Claim | Status

T1–T6, C1–C4 | **Proved** (P1-04, P1-05; proofs re-read in this audit). Computation: supporting evidence (Layers B–D).

C5 (corrected), C6 | **Proved** (P1-06). Independently re-derived symbolically (Layer A) + exact enumeration evidence.

N1 (Cov = 0), N2 (Var(dim C)), N3 (LCD refinement) | **Proved in §4 of this file** (complete proofs; inputs: P1-05 Prop. 3.1, P1-06 Lemmas 1.1/1.3, T4). Exact enumeration/field evidence agrees everywhere tested. Adoption status: **PROPOSED** — not in the frozen blueprint.

Blueprint §2 layer identity | **Proved in §2.4** (one-paragraph proof) + numeric check.

All numeric distributions/moments quoted in §6 | **Computational evidence** from code/validate_p1_14.py (exact integer/Fraction arithmetic; sympy only in Layer A as a derivation aid). Finite checks never constitute proof.

Novelty / priority vs L1, L4, L8 | **Not addressed; not claimed.** Gate G1 DEFERRED.

9. Reproducible commands (as executed for this package)

```
python3 code/validate_p1_03.py                # baseline rerun: exit 0, output byte-identical to code/validation_p1_03_output.txt
python3 code/validate_repeated_root.py  # baseline rerun: ALL PROTOTYPE CHECKS PASSED
python3 code/validate_p1_05.py        # baseline rerun: ALL P1-05 MULTI-COMPONENT ASSEMBLY CHECKS PASSED
python3 code/validate_p1_06.py        # baseline rerun: ALL P1-06 MOMENT CHECKS PASSED
python3 code/validate_p1_14.py                  # NEW: exit 0 in ~8s; full output saved as code/validation_p1_14_output.txt
                                      # final line: ALL P1-14 INDEPENDENT AUDIT CHECKS PASSED
                                      # (Layer A uses sympy if importable; Layers B–D are stdlib-only)
```

Key output lines: "Layer A done: every P1-06 closed form re-derived symbolically; gamma_self = P(P+2)/24 confirmed" · "orbit grid: 52 (P,L) pairs; … N1 (Cov=0 in 58 orbit configs incl. weights), N2 all checked" · six FIELD … blocks (§6) · "OK: P1-05 global counts reproduced independently; moments verified".